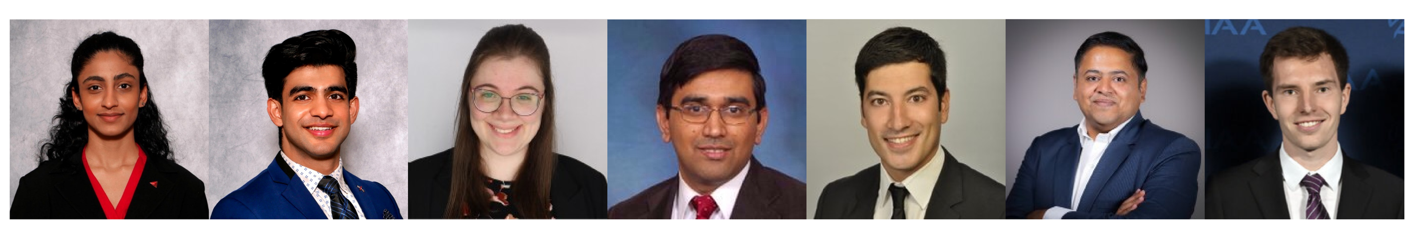
Ensuring UAS Airworthiness: Deep Learning Based Acoustic Health Monitoring of Motor Health

Abstract

This work presents an inflight UAV powertrain health monitoring framework using machine learning for real time anomaly detection. We collected high fidelity acoustic signatures from Brushless DC motors to develop a semi supervised 1D Convolutional Neural Network Autoencoder. By training exclusively on healthy acoustic profiles, the system identifies mechanical degradation by analyzing reconstruction error thresholds during flight. This non invasive approach supports preflight checks and active monitoring within the In time Aviation Safety Management System (IASMS) to ensure airworthiness in mission critical environments.

Research Team



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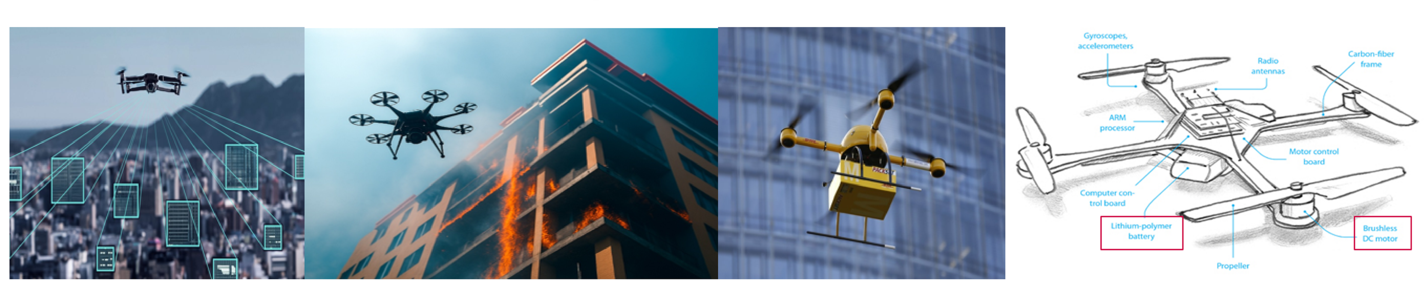
This project represents a multidisciplinary collaboration across UC, ZHAW, TU Delft, and NASA.

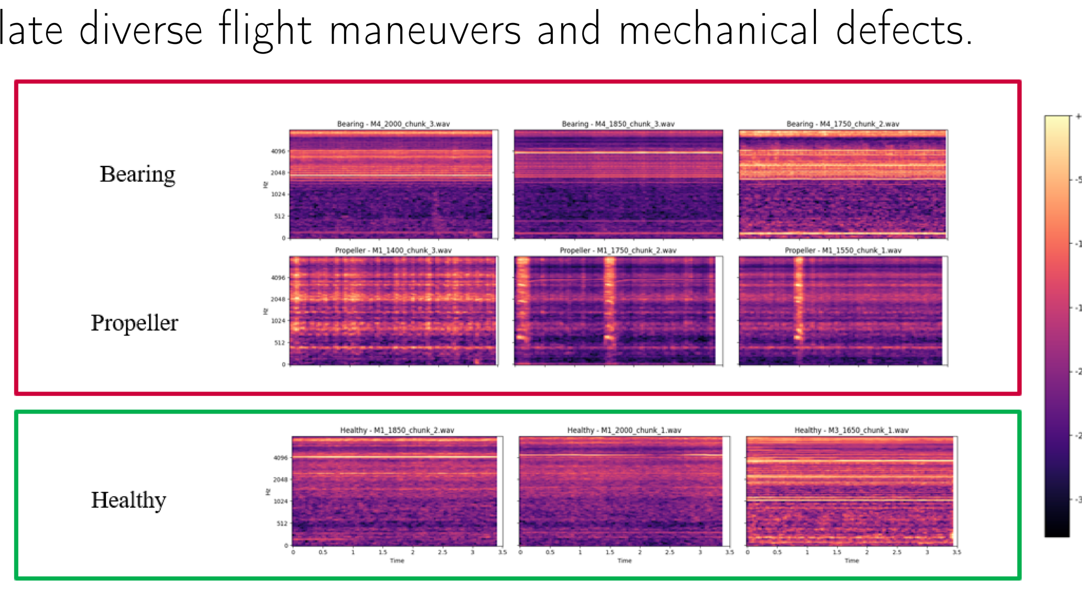
Background & Aims

Acoustic signal analysis offers an automated, non-invasive alternative to manual inspections for detecting UAV motor and propeller faults.



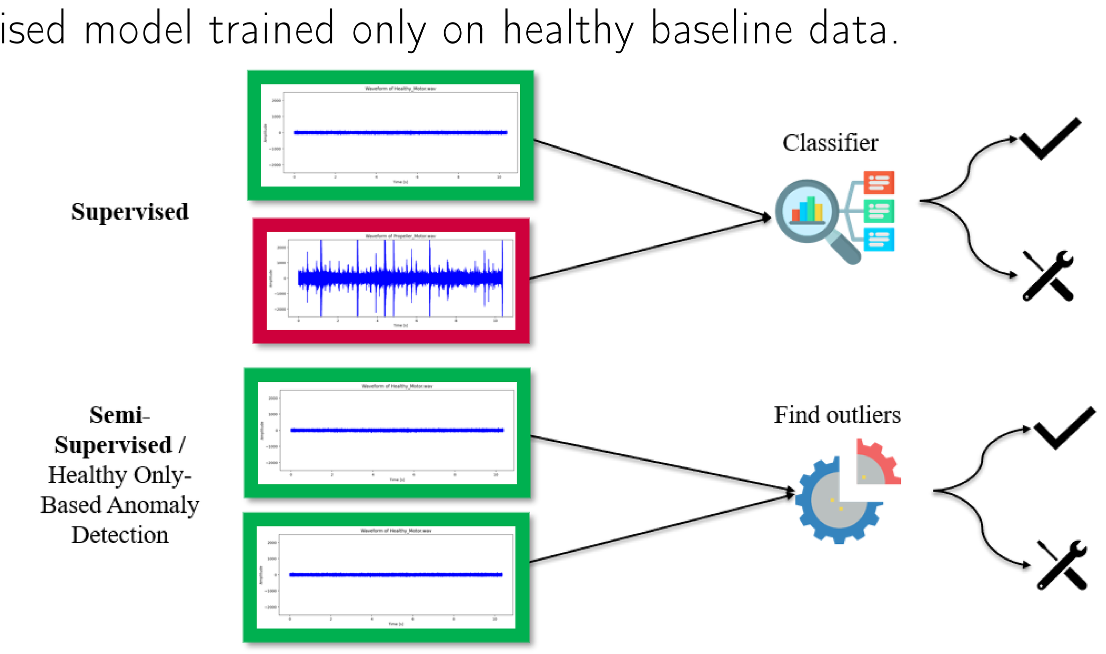
Design of Experiment

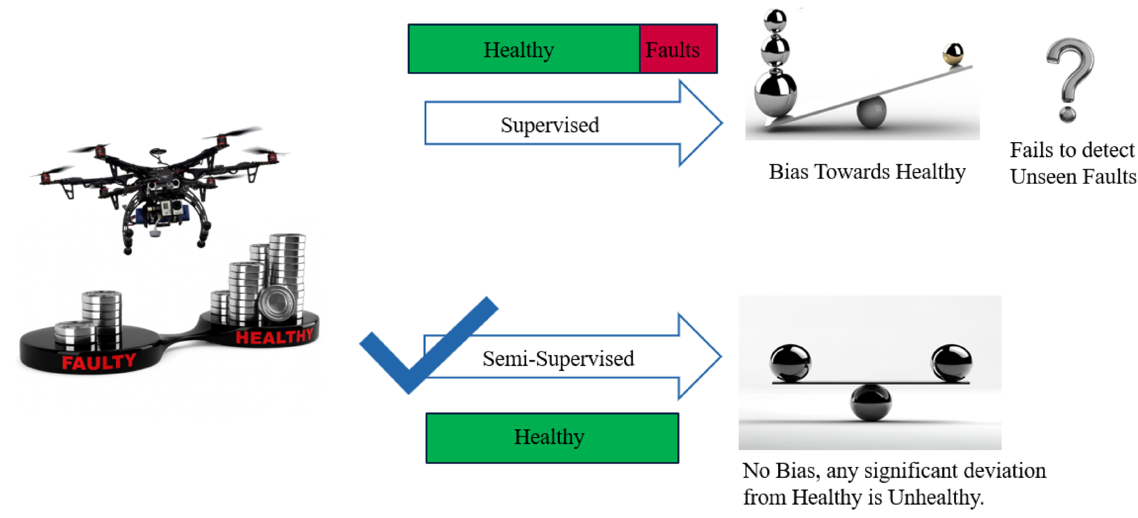
A quadcopter setup captured 477 acoustic files across 9 operational conditions to simulate diverse flight maneuvers and mechanical defects.



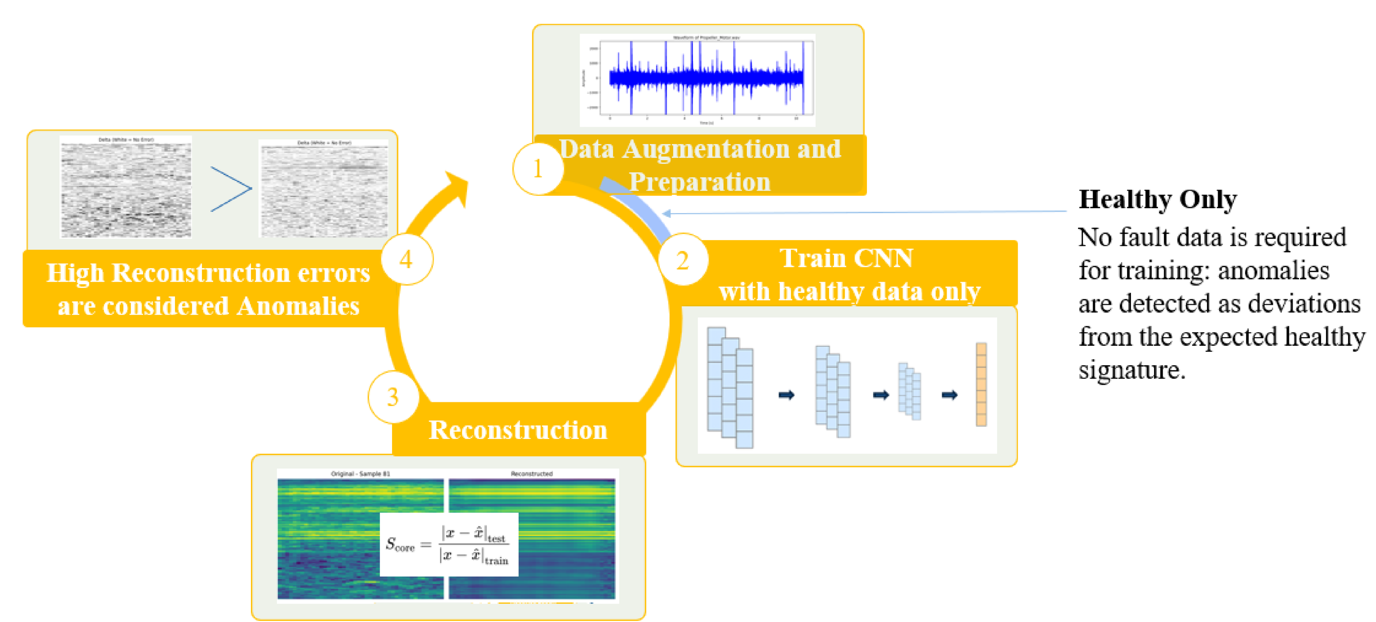
Modeling Strategy

Our approach shifts from data-heavy supervised learning to a robust semi-supervised model trained only on healthy baseline data.

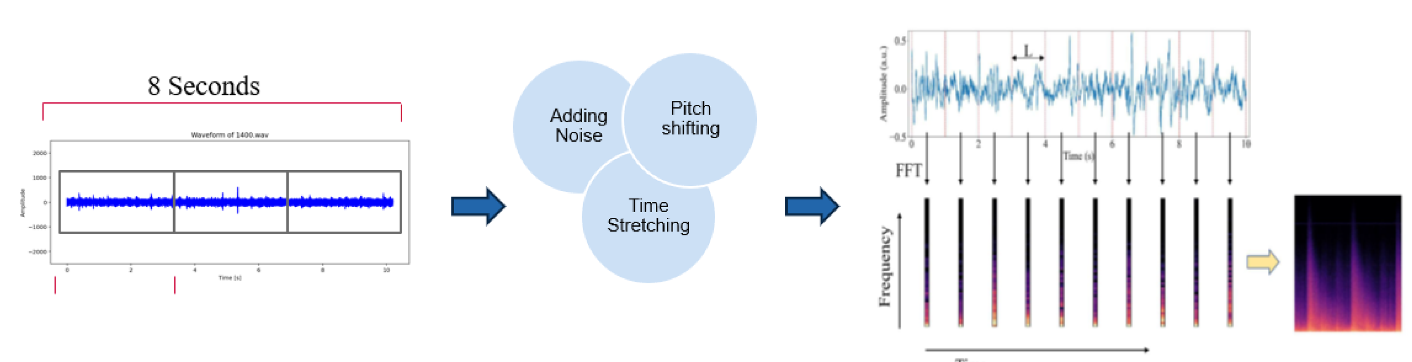




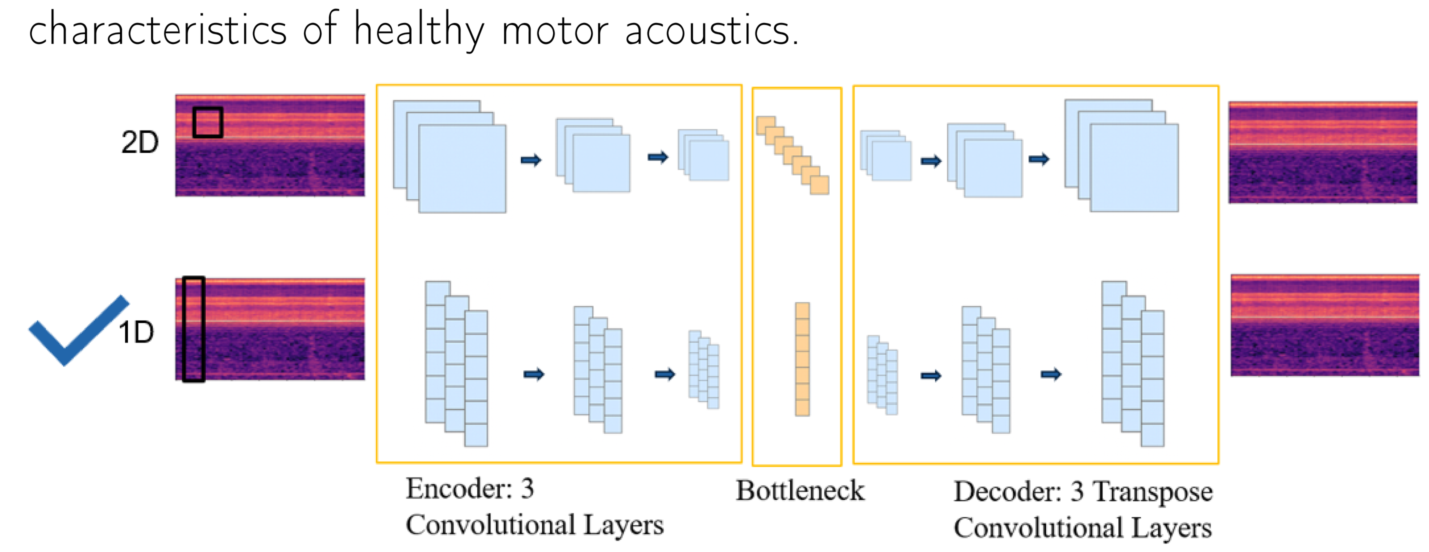
Methodology Pipeline



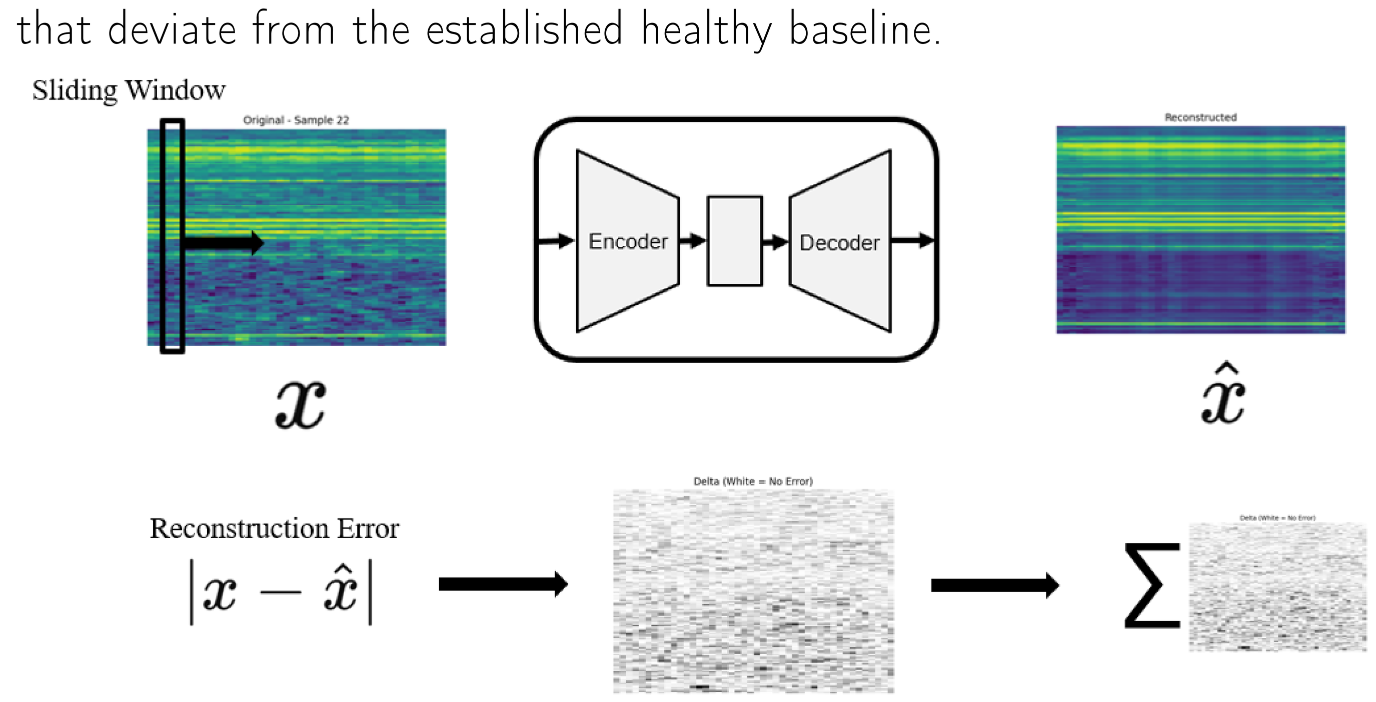
Data preprocessing involves data augmentation, normalization and converts .wav files to spectrogram images.



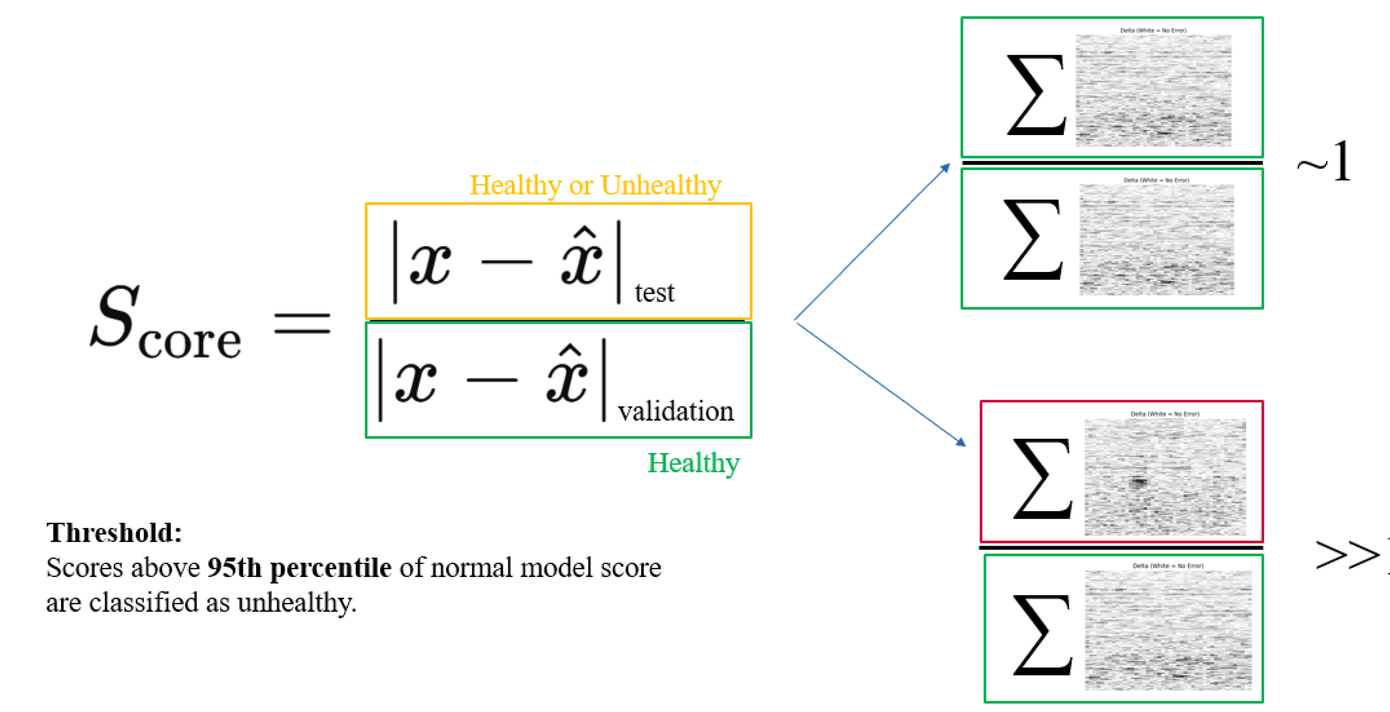
The 1D CNN Autoencoder architecture is designed to learn the latent temporal characteristics of healthy motor acoustics.



High reconstruction error serves as the primary metric for identifying signals that deviate from the established healthy baseline.

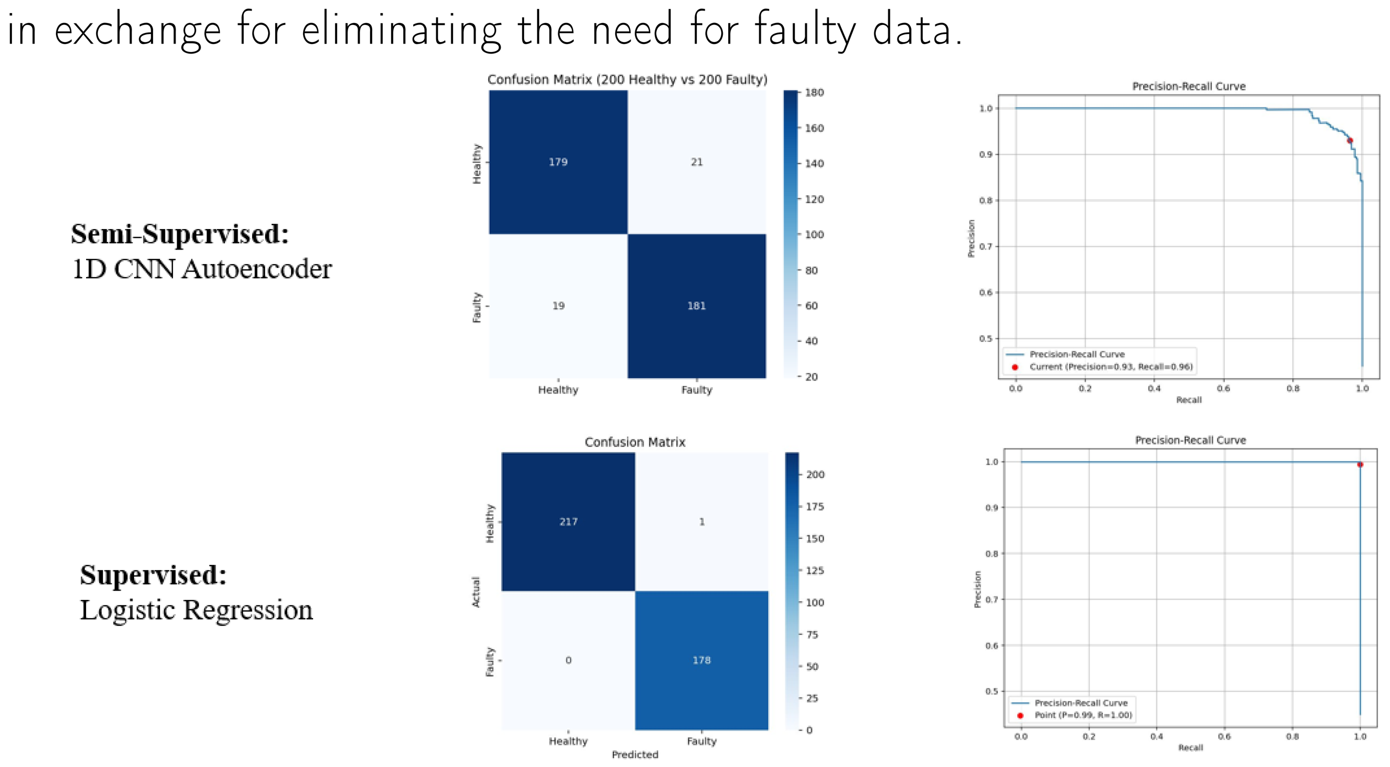


Final classification is determined by a statistical threshold, separating nominal operations from potential mechanical failures.



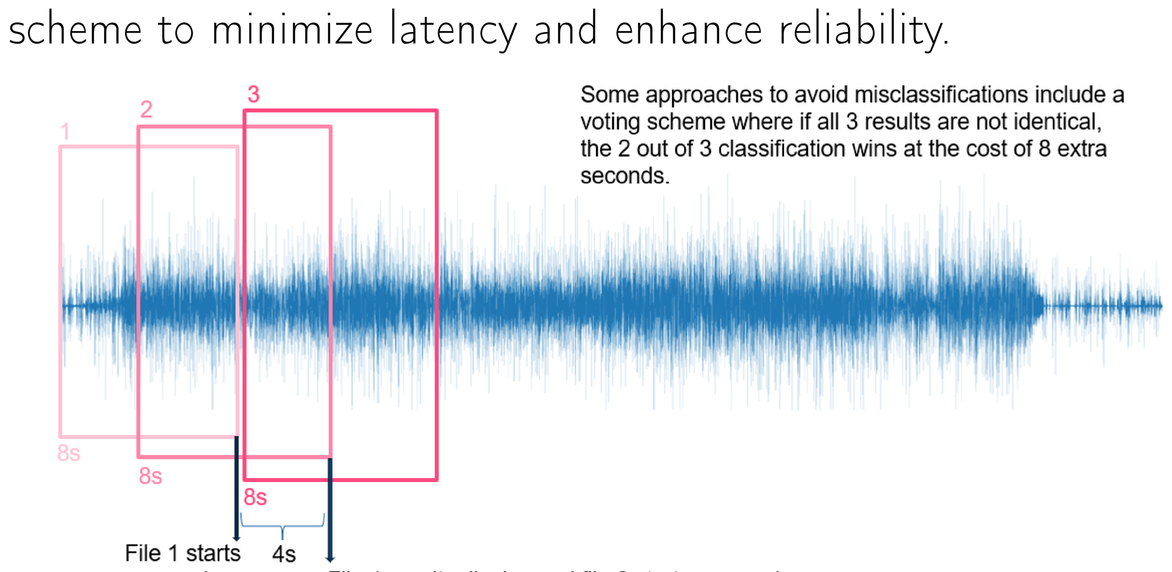
Results

A tradeoff is seen, where the autoencoder compromises slightly in performance in exchange for eliminating the need for faulty data.



Real Time Implementation

Continuous monitoring uses overlapping 8-second windows and a temporal voting scheme to minimize latency and enhance reliability.



Conclusion & Future Work

Future research will explore Transfer Learning using multiple RPM specific models or integrated models that accept RPM as a weighted input. We will also focus on conditional inference where the system triggers only during specific confidence windows such as throttle levels exceeding 50%.

References

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